

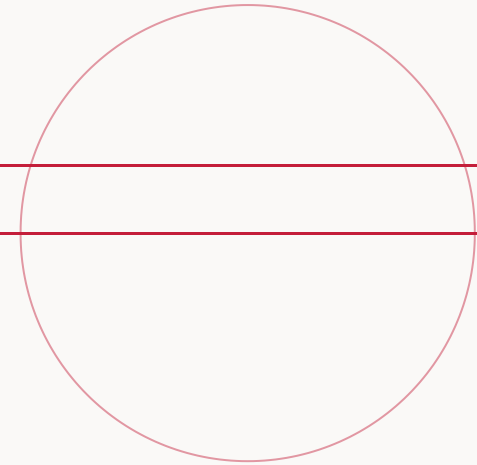
Probability Models & Parametric Risk

*Distribution fitting, target-return probability,
and Value at Risk.*

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A Course on Quantitative Finance



MOTIVATION

What can a probability model do that a histogram can't?

A histogram tells us *what did happen*. A fitted probability model lets us reason about *what could happen* — including events we haven't yet observed.

- Compute $P(R > \text{target})$ — the chance of beating a goal
- Compute $P(R < -\text{loss})$ — the chance of a crash
- Extract **quantiles** — how bad a day should we prepare for? (*Value at Risk*)
- **Extrapolate** beyond the worst day we've seen in the sample

CHAPTER I

Fitting distributions
— *two candidates*

CANDIDATE I

The normal distribution

Two parameters: mean μ and standard deviation σ . Maximum-likelihood estimates are the sample mean and sample standard deviation.

MLE ESTIMATES FOR THE NORMAL

$$\hat{\mu} = \bar{r} = \frac{1}{n} \sum_{i=1}^n r_i$$
$$\hat{\sigma} = s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2}$$

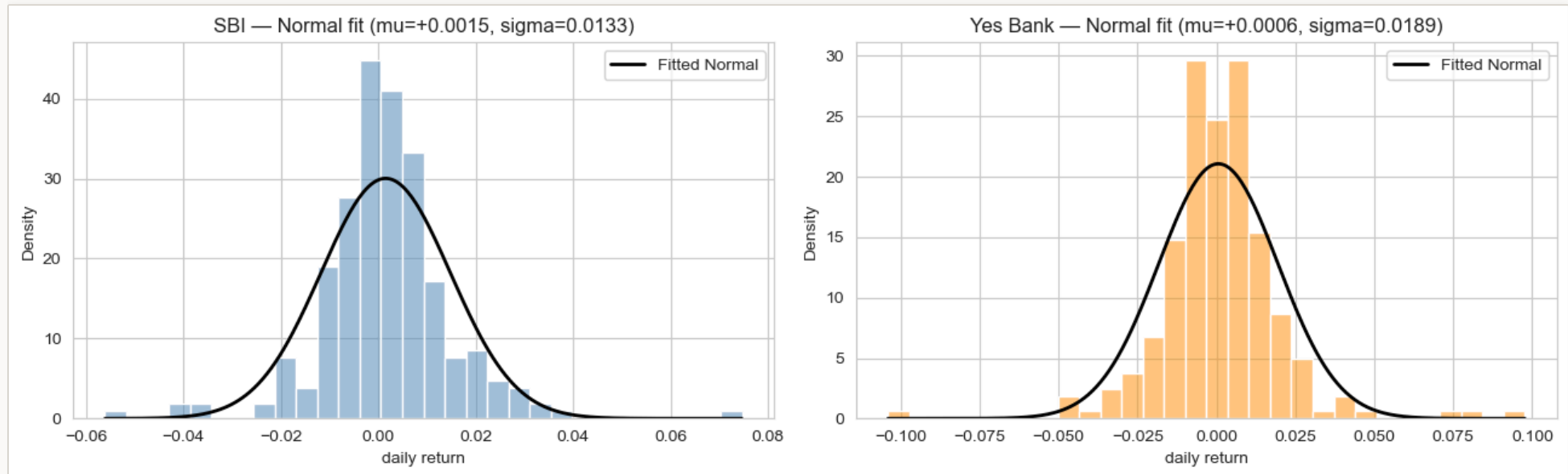
Closed-form — no iterative solver needed.

Easy to fit, but often misses the tails. *The normal under-predicts extreme moves for equity returns.*

VISUALISATION

Normal fit against empirical histograms

- ▶ The curve captures the **centre** of the distribution
- ▶ Histogram has **more mass in the tails** than the curve predicts
- ▶ A classic stylised fact of equity returns



CANDIDATE II

The Student- t distribution

Three parameters. The extra one controls **tail thickness**.

PARAMETERS OF STUDENT-T

ν (degrees of freedom), μ (location), σ (scale)

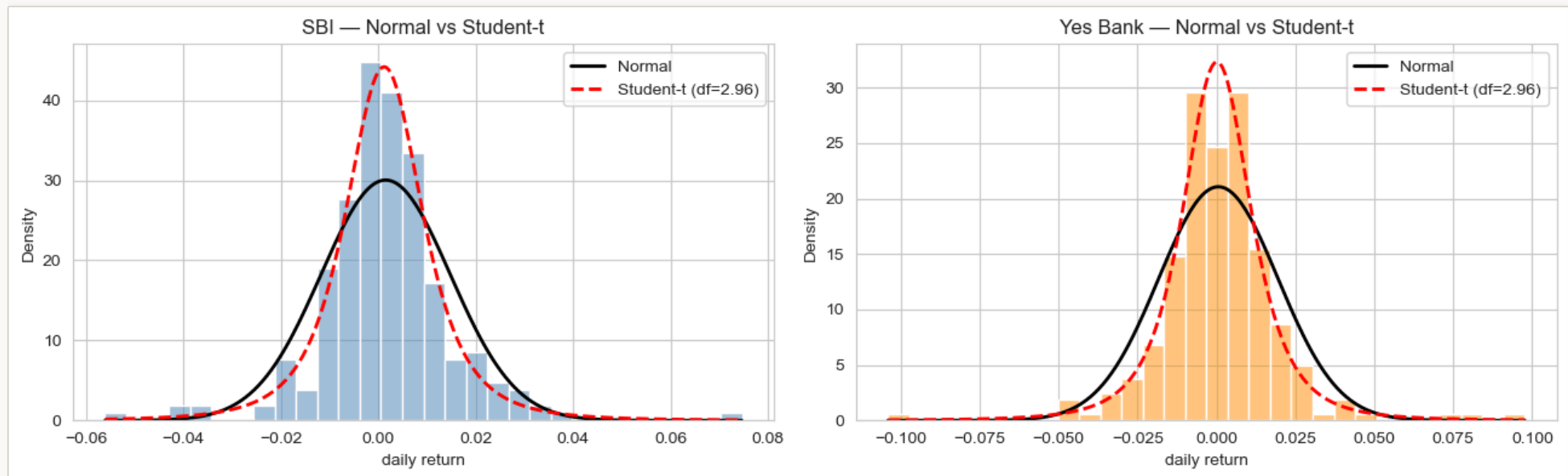
Small $\nu \rightarrow$ very fat tails. Large $\nu \rightarrow$ indistinguishable from normal. Variance is $\sigma^2 \cdot \nu / (\nu - 2)$ for $\nu > 2$.

*Fitted by **numerical MLE** — no closed form for ν . For our data the estimated ν sits around **3** — very heavy tails indeed.*

VISUAL COMPARISON

Normal vs Student- t

- ▶ Student- t is **more peaked** at the centre
- ▶ And **heavier** in the tails
- ▶ Both effects magnified when ν is small



GOODNESS OF FIT

The Kolmogorov–Smirnov test

How close is the empirical CDF to a proposed theoretical one?

The K-S statistic is the **largest vertical gap** between them.

K-S STATISTIC

$$D = \sup_x |F_n(x) - F(x)|$$

Smaller D = better fit. p -value tests H_0 : data $\sim F$.

When we've estimated parameters from the same data, the p -value is optimistic. Read it alongside the Q-Q plot — not as a sole arbiter.

GOODNESS OF FIT · RESULTS

Student-*t* demonstrably closer to the data

	NORMAL <i>D</i>	NORMAL <i>P</i>	STUDENT- <i>T</i> <i>D</i>	STUDENT- <i>T</i> <i>P</i>
<i>SBI</i>	0.0822	0.073	0.0282	0.988
<i>Yes Bank</i>	0.0921	0.031	0.0363	0.896

Both stocks — *D* shrinks roughly **3×**, *p*-values jump from ~0.05 to ~0.9.
The Student-t is the better parametric description.

CHAPTER II

Using the fitted model — two applications

QUANT APPLICATION

Probability of beating a target

Given a fitted CDF, any target return turns into a simple look-up.

TARGET-RETURN PROBABILITY

$$P(R_{\text{weekly}} > \text{target}) = 1 - F(\text{target}; \hat{\theta})$$

ϑ = the fitted parameters (μ, σ for normal; ν, μ, σ for Student-t).

Investor's question: "What's the probability my stock delivers more than 5% next week?"

RESULTS

P(weekly return > 5%)

	UNDER NORMAL	UNDER STUDENT-T
SBI	7.3%	4.9%
Yes Bank	12.9%	12.5%

The two models **disagree** — especially far out in the tail. A Student-t fit has fatter tails, so it often re-apportions mass between the centre and the extreme right.

RISK APPLICATION

Value at Risk — the loss threshold

At confidence $1 - \alpha$, Value at Risk is the loss level exceeded only with probability α .

- **95% VaR** — the loss we'd exceed on ~1 day in 20
- **99% VaR** — the loss we'd exceed on ~1 day in 100
- Reported as a **positive loss number** (*convention*)

PARAMETRIC METHOD

VaR from a fitted distribution

Once we've fitted F , VaR is just the α -quantile — expressed as a positive loss.

PARAMETRIC VALUE AT RISK

$$\text{VaR}_\alpha = -F^{-1}(\alpha; \hat{\theta})$$

Under the normal: $-(\hat{\mu} + \hat{\sigma} z_\alpha)$. Under Student-t: $-\text{ppf}(\alpha; \hat{\nu}, \hat{\mu}, \hat{\sigma})$.

RESULTS

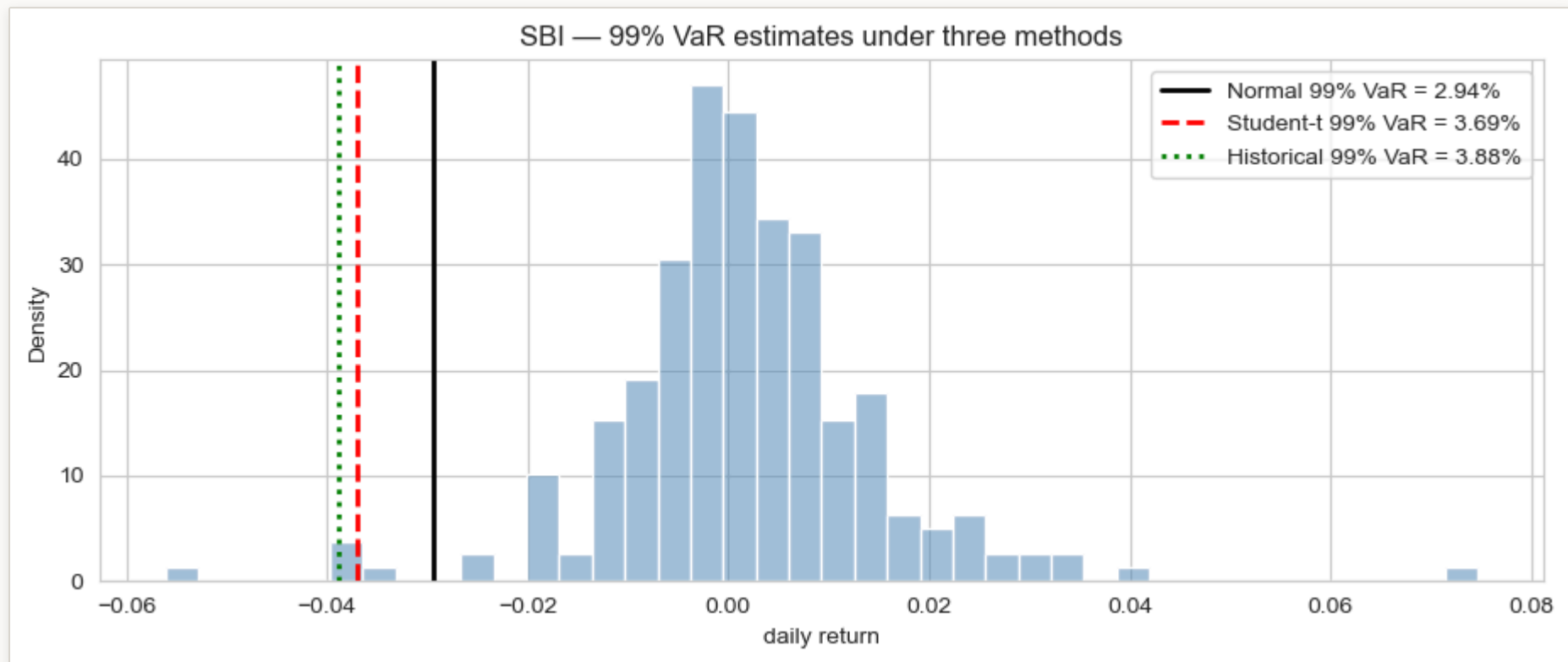
1-day VaR — three methods, two confidence levels

	NORMAL 95%	STUDENT-T 95%	NORMAL 99%	STUDENT-T 99%
SBI	2.03%	1.85%	2.94%	3.69%
Yes Bank	3.05%	2.67%	4.34%	5.19%

At 99%, Student-t VaR is **larger** than Normal — the fat-tail story translates directly into a more conservative risk estimate.

Where does each 99% VaR sit on the histogram?

- ▶ **Black solid** — Normal 99%
- ▶ **Red dashed** — Student-*t* 99%
- ▶ **Green dotted** — Historical 99%
- ▶ Student-*t* and Historical are more conservative



A THIRD METHOD · NON-PARAMETRIC

Historical VaR — no model, just data

Take the α -quantile of observed returns. That's it. No distributional assumption required.

- Pros — **assumption-free**; reflects exactly what has happened
- Cons — driven entirely by the sample; **can't extrapolate** past the worst day seen
- Useful as a **sanity check** on a parametric VaR

VALIDATION

Backtesting — did the model hold up?

A 95% VaR should be breached on ~5% of days. Count the observed breaches and compare.

BREACH RATE

$$\text{breach rate} = \frac{\#\{r_i < -\text{VaR}_\alpha\}}{n}$$

*Expected under the model: α . More breaches $\rightarrow$ model **under-estimates** risk.*

THE WARNING

Normal VaR under-covers at 99%

SBI 99% VAR	EXPECTED BREACHES	OBSERVED BREACHES
<i>Normal</i>	2.4	5
<i>Student-t</i>	2.4	4
<i>Historical</i>	2.4	3

Normal over-promises. At 99% we expected ~2 bad days in the year — we got **5**. Student-t and Historical are noticeably closer to calibrated.

CLOSING

Four things to remember

- A probability model unlocks questions a **histogram cannot answer**.
- The **normal** fits the centre well, the **tails** poorly.
- The **Student- t** is usually the better fit for equity returns — the K-S test agrees.
- Parametric VaR is *powerful* but *distribution-sensitive*: at **99%**, choose with care.

All models are wrong; some are useful.

— *choose your distribution with open eyes.*